Customer Shopping Behaviour Analysis

A Data-Driven Approach to Understanding Consumer Purchase Patterns

1. Abstract

This project focuses on analysing customer shopping behaviour using transactional data from 3,900 purchases across multiple product categories. The objective is to uncover patterns in customer demographics, purchasing trends, subscription behaviour, and product performance. Using Python for data cleaning and exploratory analysis, PostgreSQL for structured business queries, and Power BI for visualization, the project provides actionable business insights to enhance revenue strategies and customer retention.

2. Introduction

Understanding customer shopping behaviour is crucial for businesses aiming to increase revenue, improve customer retention, and optimize marketing strategies. This project leverages transactional and demographic data to identify:

- Spending patterns across customer segments
- Product performance and preferences
- Impact of discounts and promotions
- Subscription behaviour trends
- Revenue contribution across age and gender groups

The insights derived from this analysis can support strategic decision-making in pricing, promotions, and customer engagement.

3. Dataset Overview

The dataset consists of transactional records capturing customer demographics and purchasing behaviour.

Dataset Summary

- **Total Rows:** 3,900
- **Total Columns:** 18
- **Missing Values:** 37 missing values in the `review_rating` column

Key Features

Customer Demographics:

- Age
- Gender
- Location
- Subscription Status

Purchase Details:

- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Colour

Shopping Behaviour:

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

4. Data Preparation and Exploratory Data Analysis (Python)

Data cleaning and preprocessing were conducted using Python (pandas library).

4.1 Data Loading

- Imported dataset using `pandas`
- Verified structure using `df.info()`
- Generated statistical summary using `df.describe()`

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

4.2 Missing Data Handling

- Identified 37 missing values in `review_rating`
- Imputed missing ratings using the **median rating per product category** to maintain category-level consistency

4.3 Column Standardization

- Converted all column names to **snake_case** format
- Improved readability and ensured SQL compatibility

4.4 Feature Engineering

To enhance analytical depth, new features were created:

- **age_group**
  - Binned customers into defined age ranges (e.g., 18–25, 26–35, 36–45, etc.)
- **purchase_frequency_days**
  - Derived from frequency-related fields to standardize behavioral comparison

4.5 Data Consistency Check

- Evaluated redundancy between `discount_applied` and `promo_code_used`
- Found high overlap between the two variables
- Dropped `promo_code_used` to prevent duplication in analysis

4.6 Database Integration

- Connected Python to **PostgreSQL**
- Uploaded cleaned dataset into the database
- Prepared structured environment for SQL-based business analysis

5. Business Analysis Using SQL (PostgreSQL)

Structured SQL queries were used to answer real-world business questions.

5.1 Revenue by Gender

- Compared total revenue generated by male vs. female customers
- Identified the dominant revenue-generating gender segment

	gender text	revenue numeric
1	Female	75191
2	Male	157890

5.2 High-Spending Discount Users

- Identified customers who:
  - Used discounts
  - Spent above the overall average purchase amount
- Helped understand whether discounts attract high-value buyers

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	99

Total rows: 839 Query complete 00:00:

5.3 Top 5 Products by Rating

- Ranked products by average review rating
- Highlighted high-satisfaction items

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

5.4 Shipping Type Comparison

- Compared average purchase amounts between:
  - Standard Shipping
  - Express Shipping
- Analysed whether faster delivery correlates with higher spending

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5.5 Subscribers vs Non-Subscribers

- Compared:
  - Average spending
  - Total revenue contribution
- Measured subscription program effectiveness

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

5.6 Discount-Dependent Products

- Identified top 5 products with the highest percentage of discounted purchases
- Helped assess margin risk areas

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

5.7 Customer Segmentation

Customers were classified into:

- **New:** Few or first-time purchases
- **Returning:** Moderate purchase history
- **Loyal:** High repeat purchase frequency

This segmentation supports personalized marketing strategies.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

5.8 Top 3 Products per Category

- Used ranking functions to identify the most purchased products within each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

5.9 Repeat Buyers & Subscription Behaviour

- Checked whether customers with more than 5 purchases are more likely to subscribe
- Helped validate subscription targeting strategies

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

5.10 Revenue by Age Group

- Calculated total revenue contribution of each age group
- Identified high-value demographic segments

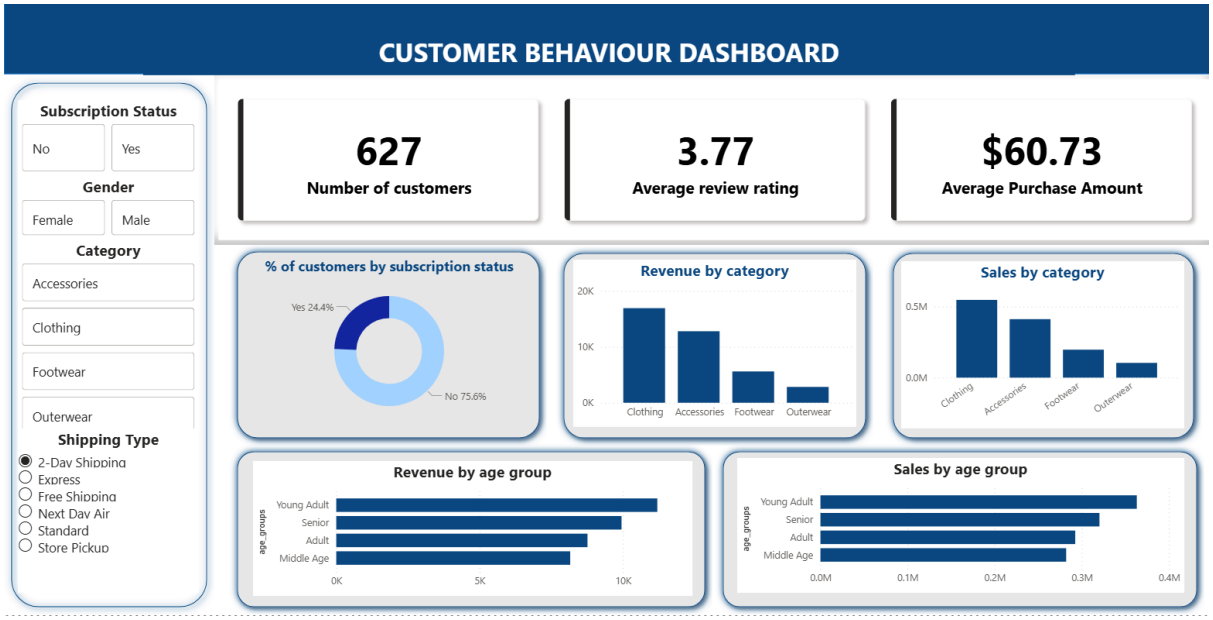
	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. Dashboard Development (Power BI)

An interactive Power BI dashboard was created to visualize:

- Revenue breakdown by gender and age group
- Subscription vs non-subscription performance
- Product category contribution
- Shipping type comparison
- Customer segmentation distribution

The dashboard allows stakeholders to filter data dynamically and make data-driven decisions.



7. Key Insights

1. Certain age groups contribute significantly higher revenue.
2. Subscribers tend to generate higher average spending.
3. Loyal customers represent a smaller segment but contribute disproportionately to total revenue.
4. Some products are heavily dependent on discounts for sales.

5. Express shipping customers tend to have higher purchase amounts.
6. Top-rated products are not always the most purchased products — indicating opportunity for better positioning.

8. Business Recommendations

8.1 Boost Subscription Programs

- Offer exclusive discounts
- Provide early access to new launches
- Implement loyalty-based subscription tiers

8.2 Strengthen Customer Loyalty Programs

- Reward repeat buyers
- Offer milestone-based rewards
- Encourage transition from Returning to Loyal segment

8.3 Optimize Discount Policy

- Reduce reliance on heavy discount-dependent products
- Introduce targeted discount strategies

8.4 Improve Product Positioning

- Promote top-rated products more aggressively
- Bundle high-rated but low-volume products

8.5 Targeted Marketing Strategy

- Focus campaigns on high-revenue age groups
- Target express-shipping users with premium offerings

9. Conclusion

This project successfully demonstrates how data analytics can transform raw transactional data into meaningful business insights. By integrating Python for preprocessing, PostgreSQL for structured analysis, and Power BI for visualization, the study provides a comprehensive understanding of customer behaviour.

The insights generated from this analysis can help businesses:

- Increase revenue
- Improve customer retention
- Optimize pricing strategies
- Design better marketing campaigns

Data-driven decision-making remains a powerful tool for gaining competitive advantage in the retail industry.